PlotTrajectory Instructions

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This is the **third step** of chemotaxis analysis in TumbleWeed. This is an optional but very helpful step for analyzing bacterial motility in a transparent way. You can either look up a single bacterium trajectory for deeper analysis or quickly validate the results from Step 2 (`CellTrackAnalysis.m`) by showing the tracking results of 20 random bacteria.

- Input: `SimpleTrackingoutput.mat` (Output file from Step 1, `SimpleTracking.m`)
- Outputs
    - Visualizations
        - `bacteria_idx =0:` Displays trajectories and tumbles of 20 random bacteria
            - This is great for quick sanity check
        - `bacteria_idx = any integer:` Input the track ID of the bacterium in interest. This will show two visualizations:
            - 1. The trajectory and tumbles display of the bacterium
            - 2. Linear and Angular speed and their local minima and maxima

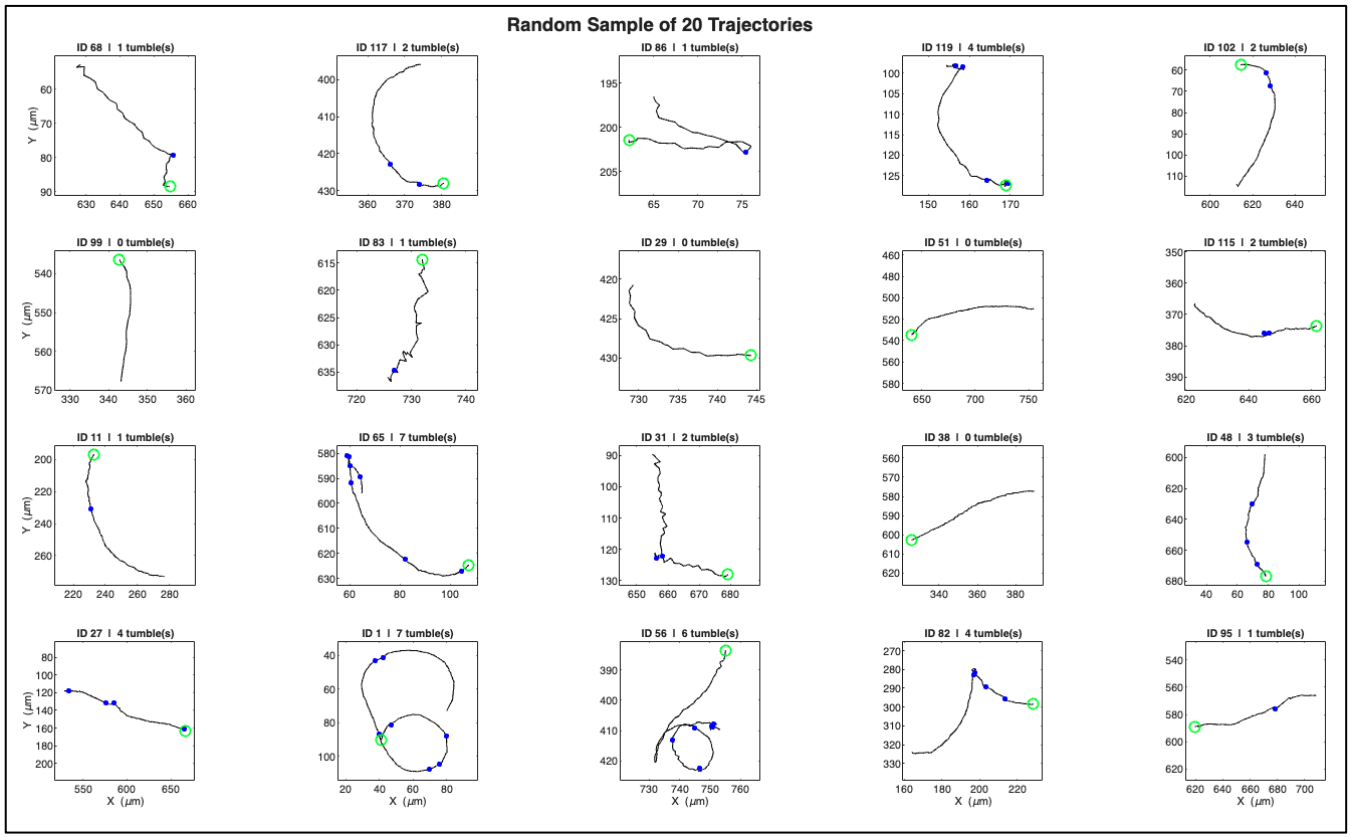
How to use PlotTrajectory

1. Open MATLAB, click "Browse For Folder" icon to navigate to TumbleWeed folder
2. Find and open `PlotTrajectory.m`
3. Adjust parameters.

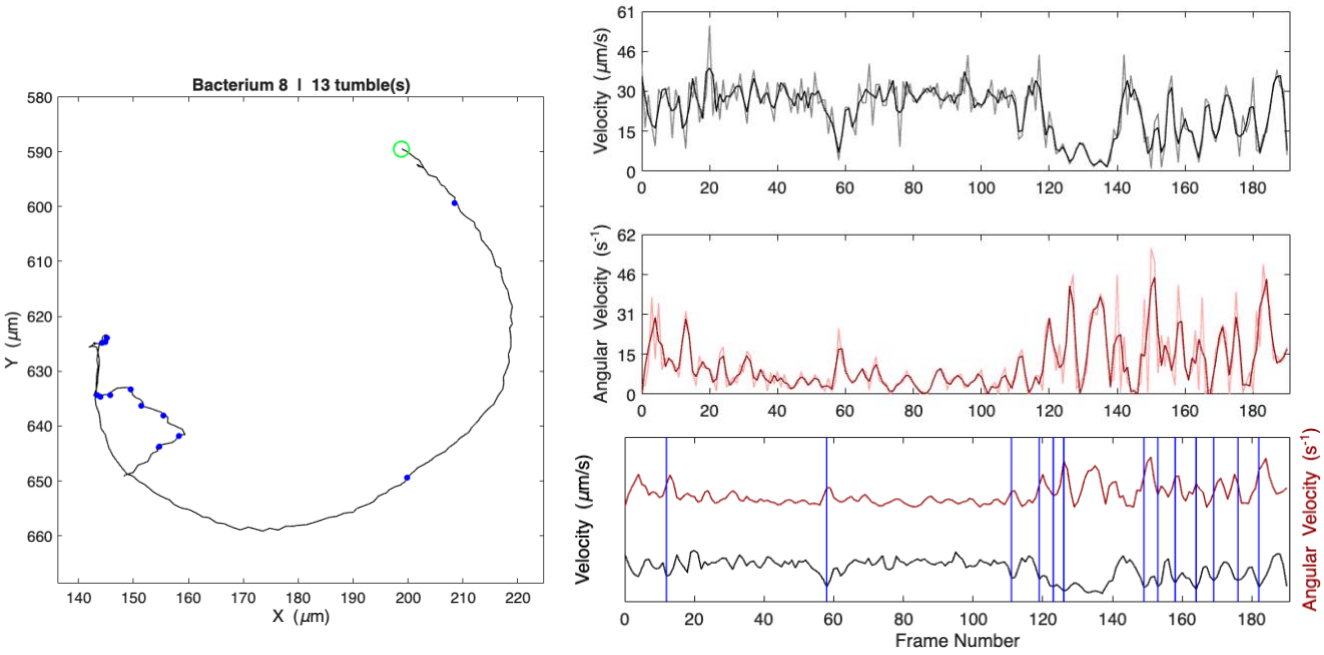
Parameter Name	Value Range	Data Type	Description
`bacteria_idx`	>= 0	Integer	**When `bacteria_idx` > 0:** The ID of the bacterium you want to look up. You can use TrackID column from `CellTrackAnalysis.xlsx`, or the ID number displayed in the scatterplot after running `CellTrackAnalysis.m`. **When `bacteria_idx` = 0:** Shows random 20 bacteria tracks.
`secondVideo`	> 0	Integer	Duration of the video in second.
`raw_mat`		String	Change the .mat file name if necessary. Default value is `SimpleTrackingoutput.mat` (Output file from Step 1)

How to interpret Results

1. `bacteria_idx = 0`
    a. This is a quick way to see if the parameters set in Step 2 (`CellTrackAnalysis.m`) works well for your data. This will show the tracks with tumbles (blue dots) of 20 random bacteria, along with track ID.
    b. In a glance, see if the setting is catching too many tumbles or too few tumbles. In that case, go back to Step 2 (`CellTrackAnalysis.m`) and try adjusting:
        i. Too many tumbles (too sensitive):
            1. Increase `depth_speed_ratio`
            2. Set `fps_scale_sg_window=false` and increase `sg_window` by 2
        ii. Too few tumbles (too conservative):
            1. Decrease `depth_speed_ratio`
            2. Set `fps_scale_sg_window=false` and decrease `sg_window` by 2
    c. Sometimes, the tracking algorithm cannot filter out noises completely and straight trajectory can look like little zigzag (like ID 31 in the figure below). If any bacterium looks suspicious, you can input that ID and look deeper into that specific bacterium.



2. `bacteria_idx > 0`
    a. Input a specific bacterium's ID and see if the tumble calculation is accurate. Great for looking at outliers or specific data points.
    b. The algorithm looks for local minima of linear speed (top right graph, black lines) and local maxima of angular speed (center right graph, red lines) and detects tumble from those points (bottom right graph). The blue vertical lines refer to the blue dots on the trajectory (left).
    c. Think as when a bacterium is "driving" and it will go straight without any direction changes if there are no obstacles (linear speed → HIGH!). However, when there are objects on the road you will change directions (angular speed → HIGH!) to avoid them and therefore your linear speed will be slow as well.



Reference

- Johnson, S., Freedman, B. & Tang, J. X. Run-and-tumble kinematics of *Enterobacter* Sp. SM3. *Phys. Rev. E* **109**, 064402 (2024).
- Cell Track Analysis Algorithm GitHub: https://github.com/Tang-BioPhys/SM3-Motility/tree/main